Name ____ Date ___________ Period _____

Domain and Range: All Three Notations

TEACHER KEY | Algebra II | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write the domain and range of a function from a graph, a table, or a set of ordered pairs using a verbal description, inequality notation, set notation, and interval notation.

Vocabulary

Domain — the set of all **input** values, the **x**-values.

Range — the set of all **output** values, the **y**-values.

Set notation — uses **braces { }** with the bar | , read "such that."

Interval notation — uses **[ ]** if an endpoint is included, **( )** if it is excluded.

Continuous — the graph is **unbroken**, with no breaks or gaps.

Discrete — the graph is made of **separate** points.

Part 1 — From a Set of Ordered Pairs

Least to greatest, and each value written only once.

Example 1. { (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) }

Domain: { **-4, 0, 2, 3, 7** }

Range: { **-7, -6, -1, 4, 7** }

You Try 1. { (-5, 3), (-1, 0), (2, 3), (6, -8) }

Domain: { **-5, -1, 2, 6** }

Range: { **-8, 0, 3** }

Part 2 — From a Table

Example 2.

x	-3	-1	0	2	5
f(x)	9	1	0	4	25

Domain: { **-3, -1, 0, 2, 5** } Range: { **0, 1, 4, 9, 25** }

You Try 2.

x	-2	0	1	4
f(x)	7	7	-3	10

Domain: { **-2, 0, 1, 4** } Range: { **-3, 7, 10** }

Part 3 — The Three Notations

The first row is done for you. Fill in the rest.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	$-\infty < x < \infty$	$\{ x \mid x \in \mathbb{R} \}$	$(-\infty, \infty)$
all reals greater than or equal to 0	$x \geq 0$	$\{ x \mid x \in \mathbb{R}, x \geq 0 \}$	$[0, \infty)$

all reals less than 5	$x < 5$	**$\{ x \mid x \in \mathbb{R}, x < 5 \}$**	**$(-\infty, 5)$**
all reals from -3 to 3, including both	$-3 \le x \le 3$	**$\{ x \mid x \in \mathbb{R}, -3 \le x \le 3 \}$**	**$[-3, 3]$**
just the numbers -2, 0, and 4	*(none)*	**$\{ -2, 0, 4 \}$**	**does not apply**

Key idea: bracket [ ] = endpoint **included**. Parenthesis ( ) = endpoint **excluded**. ∞ always takes a **parenthesis**.

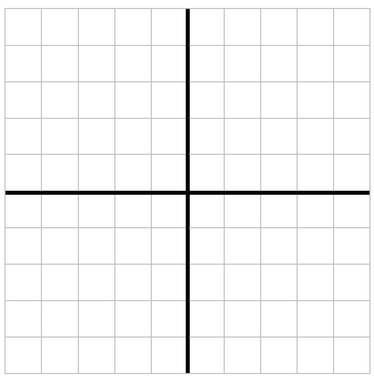
You Try 3. all reals greater than -8 → inequality **$x > -8$** interval **$(-8, \infty)$**

You Try 4. $y \le 12$ → in words **all reals less than or equal to 12** interval **$(-\infty, 12]$**

Part 4 — From a Graph

Each square is 1 unit. The axes are the two heavy lines, and both grids run -5 to 5.

A. Discrete. Plot (-4, 2), (-2, -3), (1, 4), (3, -1).

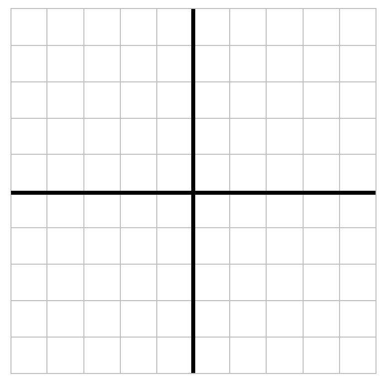


Domain (set): **$\{ -4, -2, 1, 3 \}$**

Range (set): **$\{ -3, -1, 2, 4 \}$**

Continuous or discrete? **discrete**

B. Continuous. Graph $y = x + 1$ from $x = -4$ to $x = 3$.



Domain (inequality): **$-4 \le x \le 3$**

Domain (interval): **$[-4, 3]$**

Range (interval): **$[-3, 4]$**

Part 5 — Continuous or Discrete?

Write continuous or discrete.

1. The number of students in each class period **discrete**
2. The outside temperature over one full day **continuous**
3. The total cost of n concert tickets at $45 each **discrete**

Exit Ticket

Given { (1, 5), (3, 5), (4, -2) } write the domain and range in set notation, then say whether interval notation could be used.

Domain **$\{ 1, 3, 4 \}$** Range **$\{ -2, 5 \}$**

Could interval notation be used? **no - the relation is discrete**